



Rikin Pithadia

Roll No : 23035010453

B.Sc Hons. - Data Science and Artificial Intelligence

Indian Institute Of Technology, Guwahati

d.pithadia@op.iitg.ac.in

+91-6353865443

rikinpithadia98@gmail.com

github.com/rikin-2911

linkedin.com/in/rikin-pithadia

rikin2911-portfolio-ai-ml.vercel.app

EDUCATION

Degree/Certificate	Institute/Board	CGPA	Year
B.Sc Hons. in Data Science and AI	Indian Institute of Technology, Guwahati	7.65	2023 - Ongoing
B.Tech in Mechanical Engineering	Government Engineering College, Gandhinagar	8.52	2023 - Ongoing
Minor in Internet of Things (IoT)	Government Engineering College, Gandhinagar	9.00	2024 - Ongoing

EXPERIENCE

Research Intern - SRIP'26, IIT Gandhinagar

May 2026 - Ongoing

Research under Dr. Shubhangi Bansode, Mechanical Department, IITGN

On-Site

- Developing **Scientific Machine Learning (SciML)** frameworks for modeling parametric reaction-diffusion dynamics using **Neural Ordinary Differential Equations (Neural ODEs)** and continuous-time deep learning architectures.
- Working with **Cantera-based simulation pipelines**, numerical ODE solvers, and **optimization techniques** to generate and analyze thermodynamics and chemical kinetics-driven system trajectories.
- Researching** and evaluating **Physics-Informed Neural Networks (PINNs) in PyTorch** to drive scientific advancement in the efficient modeling of nonlinear systems.

Robotics and AI Intern

Feb 2026 - May 2026

Early stage robotics and AI Start-Up

On-Site

- Engineered **embedded C/C++** robotics systems via **Arduino UNO R3, ESP32 and Raspberry Pi**, solving real-world automation challenges.
- Trained and optimized **YOLOv12-based Computer Vision models** on custom datasets for real-time object detection and deployment within robotics perception and control pipelines.
- Worked on **sensor fusion, embedded AI inference, and intelligent decision-making systems**, integrating deep learning and computer vision models with real-time robotic control architectures.

AI Engineer Intern

Aug 2025 - Nov 2025

Mirae Flux Labs Pvt. Ltd.(CognifyEV)

Remote

- Researched, implemented and optimized **RAG-based workflows and knowledge-retrieval pipelines**, improving response accuracy and system reliability across multiple internal AI applications.
- Assisted in building **Agentic AI workflows** and contributed to the development of **Python FastAPI backend services**, including API design, integration, and deployment, resulting in smoother model-backend communication.
- Collaborated with engineering teams using **Git/GitHub, code reviews, branching strategies, and issue tracking**, ensuring seamless development workflows and consistent delivery of RAG and agent automation features.

PROJECTS

1. Semantic Document Q&A & Summarization Assistant | LangChain, RAG, OpenAI

July 2025 - Aug 2025

DocuMind - "Ask. Understand. Summarize"

github.com/rikin-2911/DocuMind

- Developed **DocuMind**, an intelligent **Retrieval-Augmented Generation(RAG)**-based document assistant that enables precise **Q&A and summarization** using information retrieval pipeline to process large-scale, unstructured document datasets using various **OCR** techniques.
- Utilized **LangChain Orchestration, FAISS vector store, HuggingFace All-MiniLM-L6-V2 embeddings** and **OpenAI LLM** for semantic search, with document chunking for enhanced **context retrieval**, and deployed via a clean frontend using **Streamlit UI** for real-time, context-aware interactions.

2. Natural Image Classification | CNNs, ResNets, PyTorch, Intel Images

May 2025 - June 2025

Image Classification using Deep Learning

github.com/rikin-2911/ResNet9

- Developed a custom **ResNet9 CNN model** using **PyTorch** to classify natural scene images into **6 categories**, trained on 14,000+ images and obtained **90.27% accuracy** on validation set(3,000+ images) using advanced **image augmentation** techniques(**Resize, Random-Crop, Random-Rotation, Normalization**)
- Automated dataset retrieval via **Kaggle API** and deployed the model with a **Streamlit web app** for real-time image classification and can be utilized in **Terrain Assessment Systems**.

3. Generative AI for Text Generation | LSTM, Deepseek-R1, Streamlit

Feb 2025 - March 2025

Hack the Spring - Technoverse, GEC Gandhinagar

github.com/rikin-2911/TextGenAI

- Built a Generative AI system using **custom LSTM model** for story and dialogue generation, and integrated **Deepseek-R1 LLM via Ollama API** for a generalized **chatbot**, ranked in **top 1%** at GEC Gandhinagar Hackathon.
- Deployed both models through an interactive Streamlit web app. Demonstrated **end-to-end development**, from data preparation to UI deployment, highlighting model versatility and user-centric design.

4. IoT-Based Smart Home Energy Prediction | XGBoost, IoT Data

Jan 2025 - Feb 2025

Integration of AI/ML Systems in IoT

github.com/rikin-2911/IoTEnergyPred

- Designed an **ML pipeline** to predict energy usage in IoT-enabled smart homes using **XGBoost**, and achieved **RMSE score of 0.158** on test data.
- Achieved **97.76% accuracy** on test data through **data preprocessing (imputation, scaling, encoding)** and handled class imbalance effectively.
- Enabled **energy pattern insights** via detailed **EDA** and visualizations using **Seaborn and Matplotlib**.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Java, R, Linux Scripting
 - **Python Tools:** Numpy, Pandas, Scikit-learn, Matplotlib
 - **Deep Learning Frameworks:** PyTorch, TensorFlow, Keras
 - **Generative AI and Agentic AI:** LangChain, LangGraph, RAG, MCP, LLM Inferencing, HuggingFace models, FAISS, Pinecone
- **Machine Learning:** Supervised and Unsupervised Learning, Algorithms, Feature Engineering
 - **Deep Learning:** Computer Vision, NLP, CNN, RNN, LSTM, Transformers
 - **Deployment:** FastAPI, Docker, Streamlit, Git, GitHub

RELEVANT COURSEWORK

- **Mathematics:** Probability and Statistics, Calculus, Linear Algebra, Optimization, Statistical Inferencing
- **AI/ML:** Machine Learning Fundamentals, Deep Learning for Computer Vision, Hardware-Aware Deep Learning, Large Language Model Application Engineering, Recommender Systems
- **Electrical and Electronics:** Signals and Systems, Applied Machine Learning for Signal Processing, AI Based Wireless Communication Systems

CERTIFICATIONS AND ACHIEVEMENTS

- **Winner IDEATHON’25 - Lead team for GenAI Domain,** GEC-Gandhinagar
 - **LangChain Essentials for Generative AI,** LangChain Web
 - **Trustworthy Generative AI,** Coursera
 - **Machine Learning and Deep Learning with Python,** Jovian
 - **Business Writing - Effective Communication,** Coursera
- 2025

2025

2024

2024

2024

POSITION OF RESPONSIBILITY

- **AI/ML Lead - CLUB IDE (Innovate Develop Engage), GEC-Gandhinagar**
 - **AI Innovation:** Conducted various technical workshops on developing deep learning models with **PyTorch** and building **Agentic AI** systems using **LangChain** and **Model Context Protocol (MCP)** frameworks. Mentored students in designing modular AI pipelines, tool integration, and real-world deployment strategies.